Predicting Loan Default Using Machine Learning

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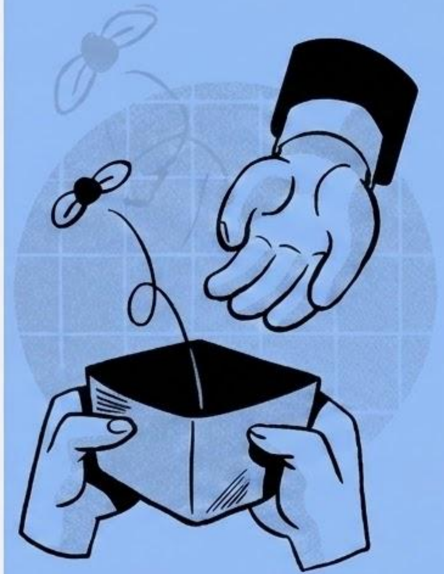
The Problem

The Challenge

Loan defaults create financial losses for lenders, making effective credit risk assessment essential.

Can machine learning identify borrowers at higher risk of default?

What factors are most associated with loan default risk?

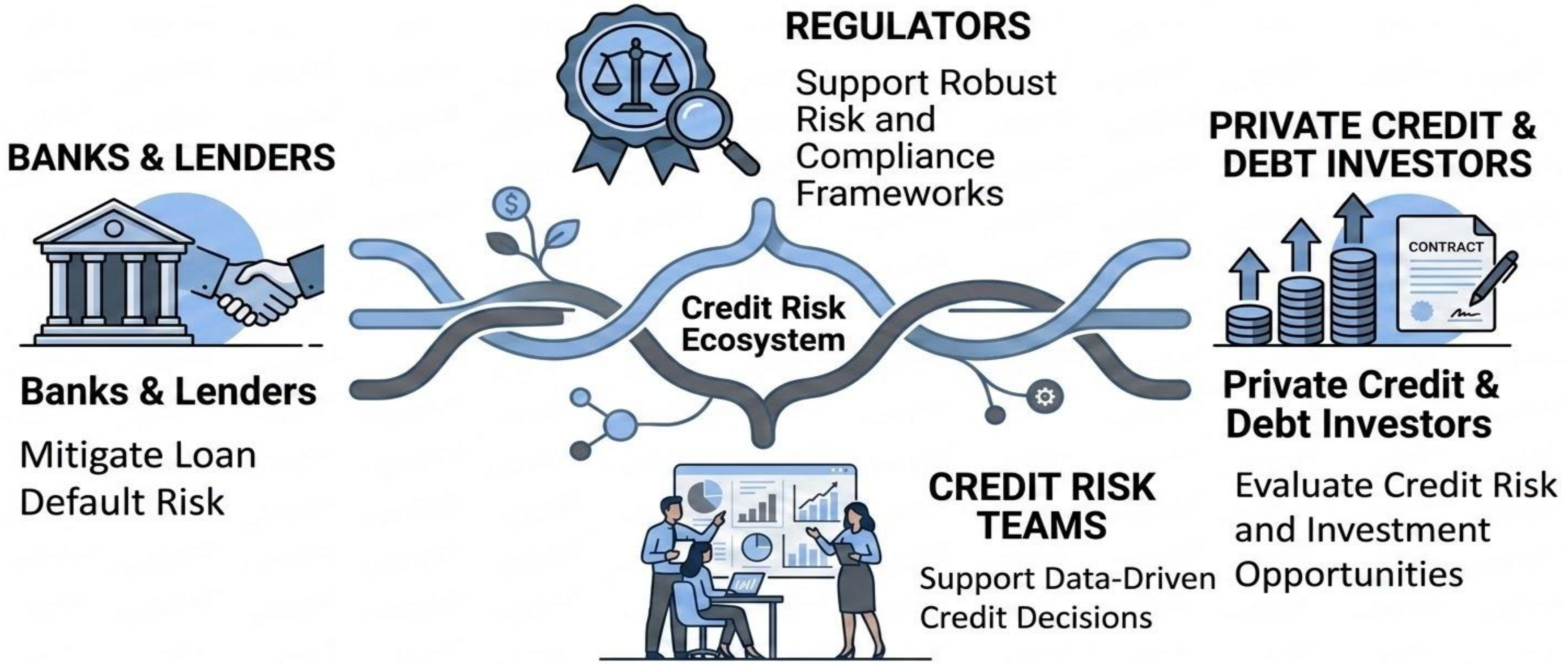


Credit Risk

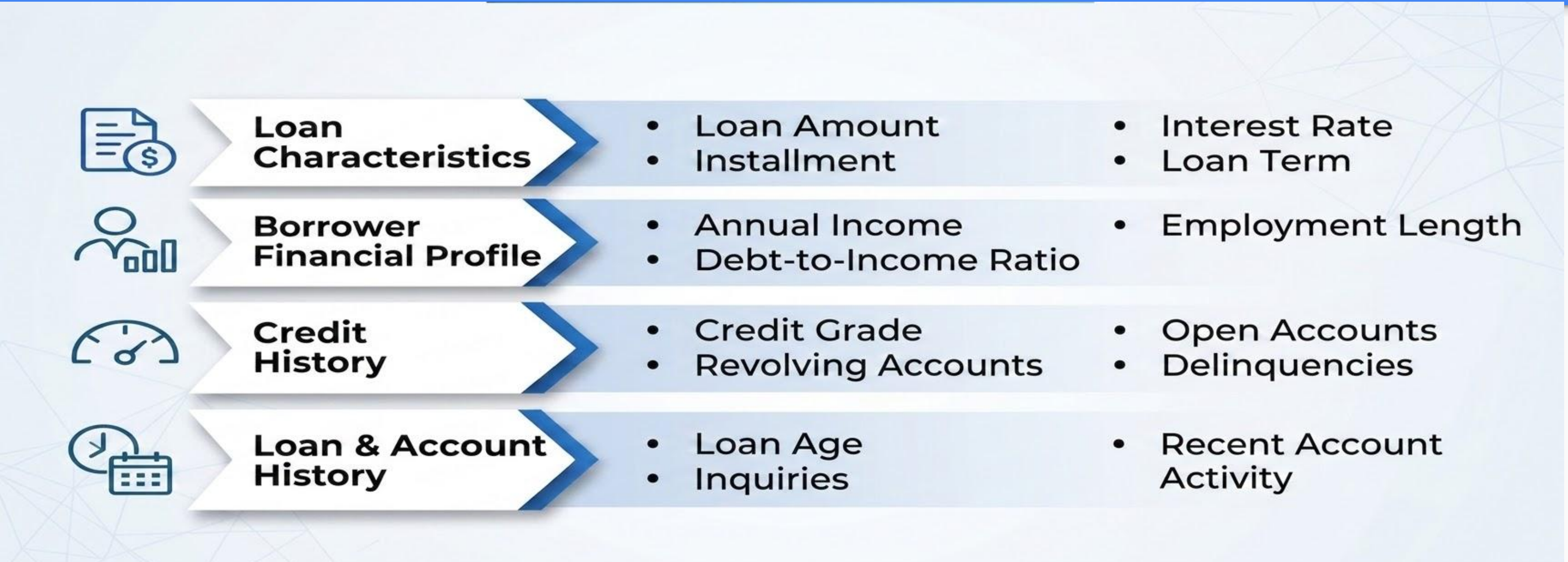
['kre-dit 'risk]

The possibility of a loss resulting from a borrower's failure to repay a loan or meet contractual obligations.

Who might care?



What factors may help identify Loan Default?



Data Information

Dataset & Source

Source Platform
Kaggle

Dataset Name
Lending Club Loan

Timeline
June 2007 - Dec 2018

Key Metrics

2.26M+
Original Loan Records

145
Original Features

75,000
Modeling Sample

Target Variable

Binary classification target classes representing repayment outcome:

0
Non-Default
Fully Paid / Current

1
Default
Charged Off / Default

Target Variable Definition

Binary Classification Problem

0–Non-Default

- Fully Paid
- Current
- In Grace Period
- Late (16-30 days)
- Late (31-120 days)
- etc

1–Default

- Charged Off
- Default
- Does not meet the credit policy. Status: Charged Off

- ❑ **This project uses supervised machine learning to solve a binary classification problem, in which the original** ‘loan_status’ **variable was grouped into two target classes based on loan repayment outcomes.**
- ❑ **The original** ‘loan_status’ **variable was removed after target creation to prevent data leakage.**

Data Preparation & Feature Transformation

01 Data Cleaning & Refinement

Removed irrelevant variables and columns with excessive missing values.

02 Leakage Prevention

Removed variables containing information unavailable at prediction time (e.g., debt_settlement_flag).

03 Missing Value Treatment

Imputed missing values and created missing-value indicator features where appropriate.

04 Feature Encoding

Converted dates into numerical features (e.g., loan_age_years).

Encoded categorical elements and derived grade_combined.

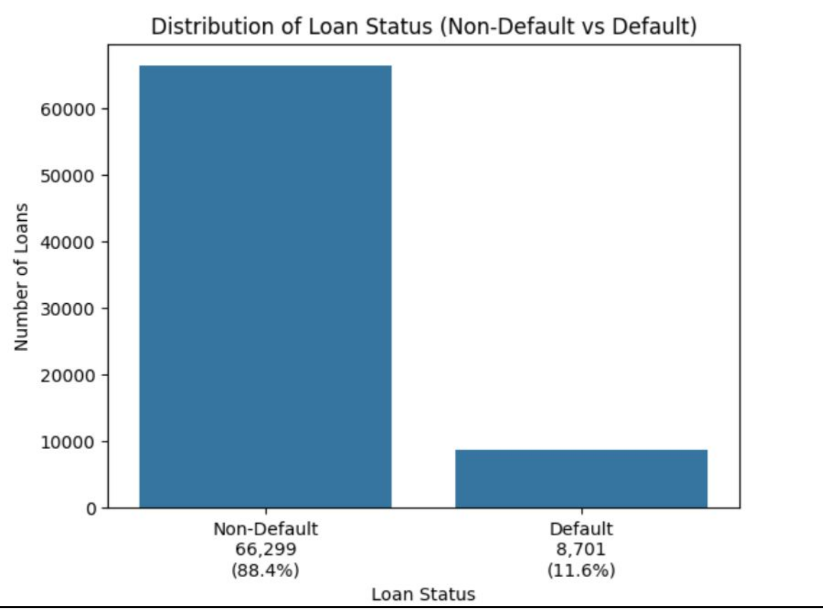
05 Model Preparation

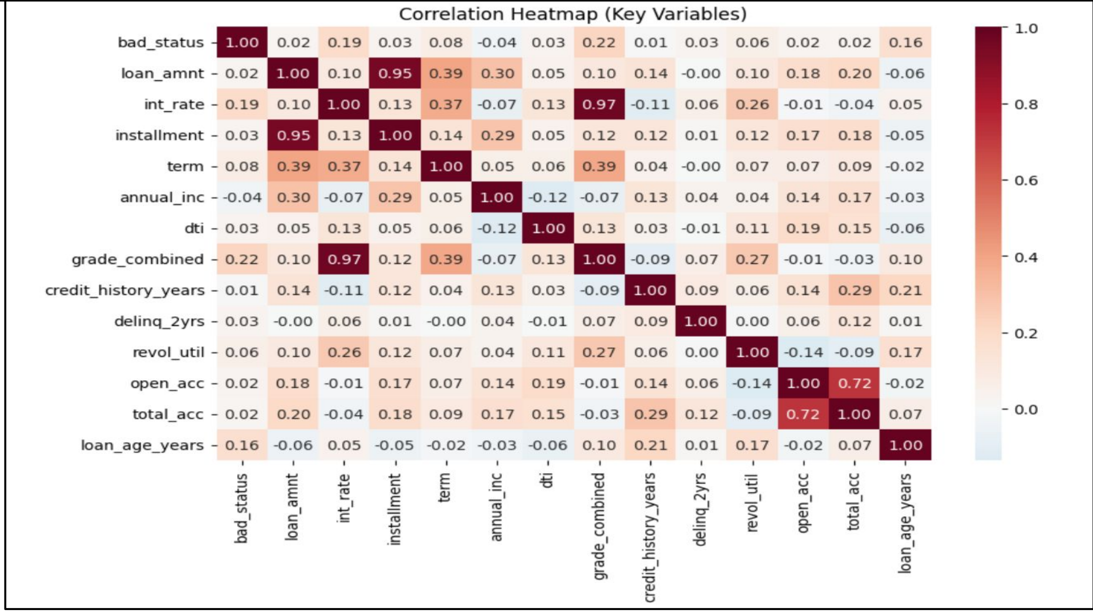
Performed standard 80/20 train-test split.

Applied robust feature scaling scross across datasets.

Exploratory Data Analysis :

Target & Correlations



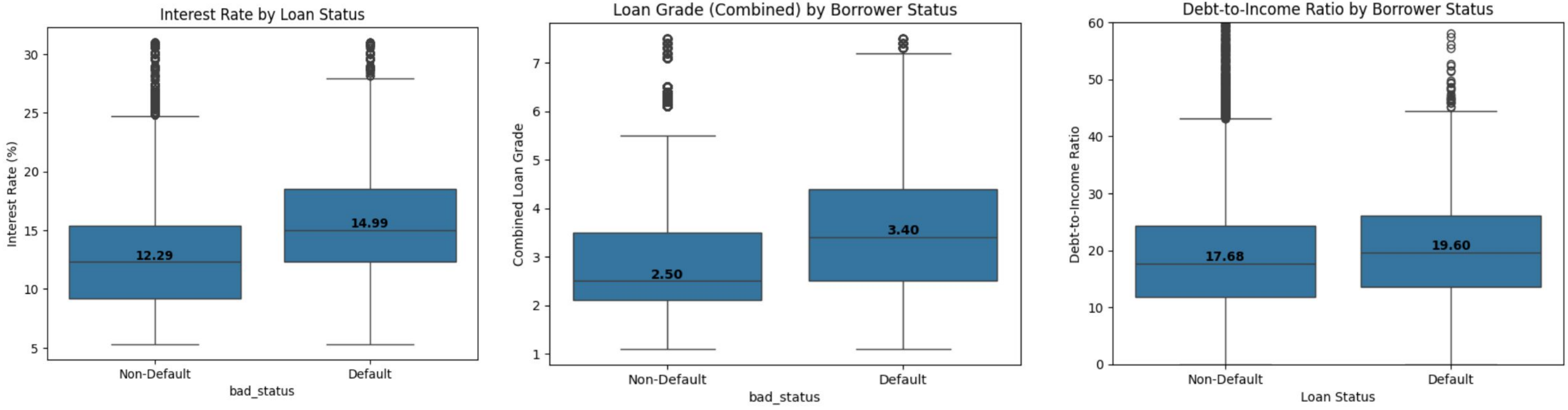


- The plot is showing a **significant class imbalance**: 88.4% non-default vs 11.6% default
- Class imbalance was addressed during modeling using techniques such as class weighting and SMOTE

- **grade_combined** showed the strongest correlation with default status (0.22) among the variables examined.
- **interest_rate** also showed a relationship with default status (0.19)
- Loan grade and interest rate were highly correlated (0.97), indicating potential multicollinearity
- Most individual features showed relatively weak correlations with default status

Exploratory Data Analysis:

Borrower Risk Characteristics



- ❑ **Interest Rate: Defaulted loans tended to have higher median interest rates (14.99% vs. 12.29%).**
- ❑ **Loan Grade: Defaulted loans tended to have higher-risk loan grades (3.40 vs. 2.50).**
- ❑ **Debt-to-Income Ratio: Defaulted loans had slightly higher DTI ratios (19.60 vs. 17.68), though the difference was relatively modest.**

Model Development

Logistic Regression

- Baseline LR
- Balanced LR
- Balanced LR with Lasso
- LR with SMOTE
- Tuned LR

Random Forest

- Baseline RF
- RF with SMOTE
- Tuned RF

XGBoost

- Baseline XGBoost
- Tuned XGBoost

MODEL OPTIMIZATION STRATEGIES

Class Weighting	Assigned greater weight to the minority default class during model training.
SMOTE Oversampling	Created synthetic minority-class samples in the training data to address class imbalance.
Hyperparameter Tuning	Tested different model configurations to optimize model performance.

Model Evaluation & Results

MODEL COMPARISON

Performance Summary of Five Models

★ Higher is better for all metrics

Evaluation Goal: Balance strong predictive performance with the ability to effectively identify defaulted loans.

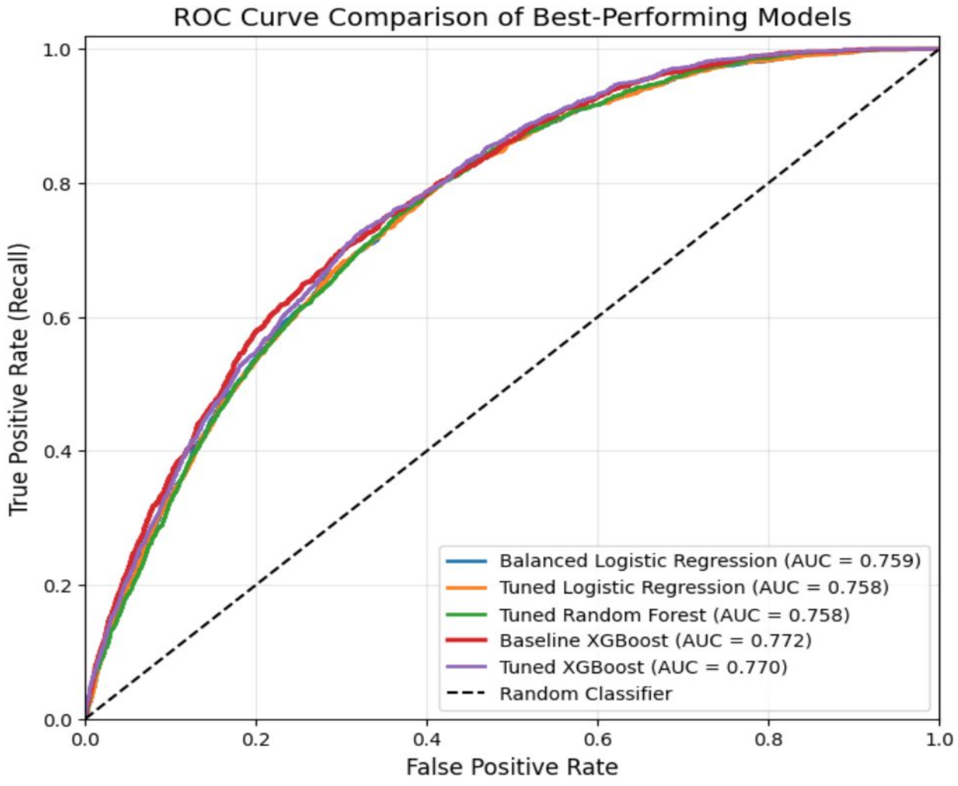
Key Takeaway: Each model offers a different trade-off between accurately identifying defaults and maintaining high overall accuracy.

METRIC	Balanced Logistic Regression	Tuned Logistic Regression	Tuned Random Forest	Baseline XGBoost	Tuned XGBoost
Accuracy	70.0%	69.0%	66.0%	69.0%	**71.0%**
Recall – Non-Default (Class 0)	70.0%	69.0%	65.0%	69.0%	**71.0%**
Recall – Default (Class 1)	68.0%	69.0%	**73.0%**	71.0%	67.0%
F1-Score: Default (Class 1)	34.0%	34.0%	33.0%	**35.0%**	**35.0%**
ROC-AUC	75.9%	75.8%	75.8%	**77.2%**	77.0%
Gini Coefficient	0.518	0.516	0.516	**0.543**	0.539

- **Balanced and Tuned Logistic Regression** provide consistent, stable performance across both classes
- **Tuned Random Forest** achieves the highest default recall (73%) but has lower overall accuracy.
- **Baseline XGBoost** offers the highest Gini (0.543) and ROC-AUC (77.2%), with strong default recall (71%).
- **Tuned XGBoost** achieves the highest accuracy (71%) and non-default recall (71%), offering strong overall performance.

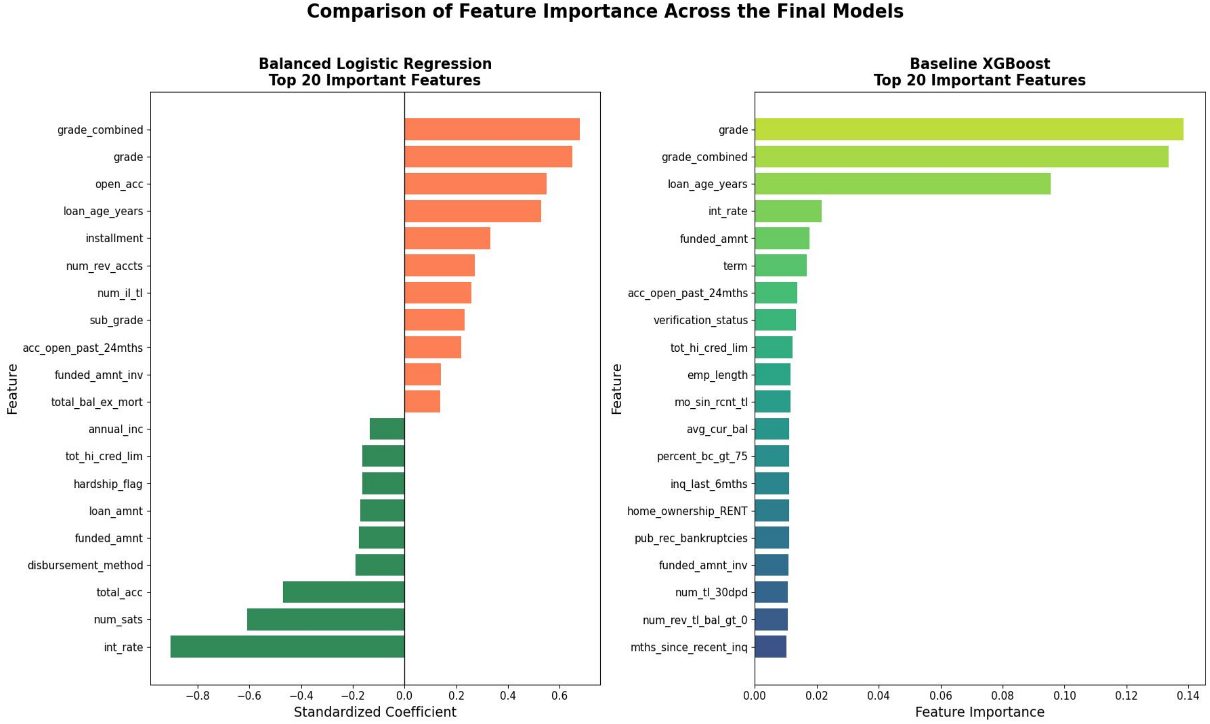
- **Accuracy:** Overall percentage of correctly classified loans; can be misleading with imbalanced data.
- **Recall (Default):** Percentage of actual defaults correctly identified — a key metric for credit risk.
- **F1-Score (Default):** Balances precision and recall for the default class.
- **ROC-AUC:** Measures the model’s overall ability to distinguish between default and non-default loans.
- **Gini:** Measures discriminatory power and is commonly used in credit-risk modeling; derived from ROC-AUC.

ROC Curve Comparison of Best Performing Models



- **All models demonstrate similar and solid discriminatory performance**, with ROC-AUC ranging from 0.758 to 0.772.
- **Baseline XGBoost** achieves the highest ROC-AUC (0.772) and Gini coefficient (0.543), with strong default recall (71%), and was selected as the final model.
- **Balanced Logistic Regression** remains a strong, interpretable alternative, with ROC-AUC of 0.759.
- **Final model choice depends on business priorities**, including default detection, overall performance, and interpretability.

Feature Importance & Key Predictors



- **Loan grade and loan age** emerged as important predictors across both models.
- **Logistic Regression** provides directional relationships, but multicollinearity among correlated features may affect coefficient magnitude and sign (e.g., interest rate)
- **XGBoost** captures more complex relationships and interactions, but feature importance does not indicate direction.
- Despite different approaches, **both models identify similar core credit-risk factors**.

Conclusions & Recommendations

Main Findings

Baseline XGBoost was selected as the final model, achieving the highest ROC-AUC (0.772) and Gini (0.543) while maintaining strong default recall (71%).

Balanced Logistic Regression remains a strong, interpretable alternative with consistent performance across both classes.

Results highlight the trade-off between default detection, overall predictive performance, and interpretability.

Scope & Application

The analysis evaluates **default risk among loans that were already originated**, rather than making loan approval or rejection decisions.

The model can support **portfolio risk assessment** by identifying patterns and characteristics associated with higher default risk.

Future Work

- Evaluate the model on the full LendingClub dataset.
- Validate performance using out-of-time data (loans issued in later years).
- Explore additional boosting algorithms, such as LightGBM and CatBoost.
- Optimize the classification threshold based on business risk priorities.
- Apply SHAP for more detailed model interpretability.